

Problem 3

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Problem 1 Suppose that $\int_1^3 f(x) dx = 4$

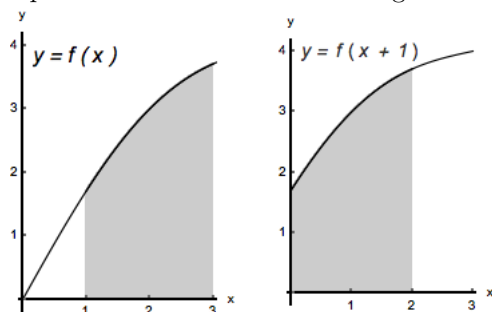
(a) Evaluate the following integrals.

(i) $\int_0^2 f(x+1) dx$

Explanation. Let $u = x + 1$. Then $du = dx$; when $x = 0$, $u = 1$; when $x = 2$, $u = 3$. Therefore,

$$\int_0^2 f(x+1) dx = \int_1^3 f(u) du = 4$$

Notice that the region whose net area is given by the integral $\int_0^2 f(x+1) dx$ is just the region whose net area is given by the integral $\int_1^3 f(x) dx$, only “shifted to the left by 1”. Obviously, their net areas have to be equal. We illustrate this in the figure below.



(ii) $\int_1^9 3 \frac{f(\sqrt{x})}{\sqrt{x}} dx$

Explanation. Let $u = \sqrt{x}$. Then $du = \frac{1}{2\sqrt{x}} dx$; when $x = 1$, $u = 1$; when $x = 9$, $u = 3$. Therefore,

$$\int_1^9 3 \frac{f(\sqrt{x})}{\sqrt{x}} dx = \int_1^9 3 \cdot 2 \cdot \frac{f(\sqrt{x})}{2 \cdot \sqrt{x}} dx = \int_1^3 6f(u) du = 6 \int_1^3 f(u) du = 6 \cdot 4 = 24$$

$$(iii) \int_{\frac{1}{3}}^1 f(3x) dx$$

Explanation. Let $u = 3x$. Then $du = 3 dx$; when $x = \frac{1}{3}$, $u = 1$; when $x = 1$, $u = 3$. Therefore,

$$\begin{aligned} \int_{\frac{1}{3}}^1 f(3x) dx &= \int_{\frac{1}{3}}^1 \frac{1}{3} f(3x) \cdot 3 dx = \int_1^3 \frac{1}{3} f(u) du = \frac{1}{3} \int_1^3 f(u) du = \frac{1}{3} \cdot 4 = \frac{4}{3} \end{aligned}$$

$$(iv) \int_2^4 3f(x-1) dx$$

Explanation. Let $u = x - 1$. Then $du = dx$; when $x = 1$, $u = 1$; when $x = 4$, $u = 3$. Therefore,

$$\int_2^4 3f(x-1) dx = \int_1^3 3f(u) du = 3 \int_1^3 f(u) du = 3 \cdot 4 = 12$$

$$(v) \int_0^{\sqrt{2}} 3xf(x^2+1) dx$$

Explanation. Let $u = x^2 + 1$. Then $du = 2x dx$; when $x = 0$, $u = 1$; when $x = \sqrt{2}$, $u = 3$. Therefore,

$$\begin{aligned} \int_0^{\sqrt{2}} 3xf(x^2+1) dx &= \int_1^3 3 \cdot \frac{1}{2} \cdot 2x f(x^2+1) dx = \int_1^3 \frac{3}{2} f(u) du = \frac{3}{2} \cdot 4 = 6 \end{aligned}$$

$$(b) \text{ Assume that } f \text{ is odd. Evaluate } \int_{-3}^{-1} f(x) dx$$

Explanation. Since f is odd, then $f(-x) = -f(x)$ or $f(x) = -f(-x)$, for all x in its domain.

Let $u = -x$. Then $du = -dx$; when $x = -3$, $u = 3$; when $x = -1$, $u = 1$.

Therefore,

$$\int_{-3}^{-1} f(x) dx = \int_3^1 [-f(-x)] (-du) = \int_3^1 f(u) du = - \int_1^3 f(u) du = -4$$

$$(c) \text{ Assume that } f \text{ is even. Evaluate } \int_{-3}^{-1} f(x) dx$$

Explanation. Since f is even, then $f(-x) = f(x)$, for all x in its domain. Let $u = -x$. Then $du = -dx$; when $x = -3$, $u = 3$; when $x = -1$, $u = 1$.

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Therefore,

$$\begin{aligned}\int_{-3}^{-1} f(x) \, dx &= \int_{-3}^{-1} [f(-x)] \, dx = - \int_{-3}^{-1} [f(-x)](-dx) = - \int_3^1 f(u) \, du = \\ \int_1^3 f(u) \, du &= 4\end{aligned}$$